Memo: Whale Position and Velocity Estimation

MANTA Simulator

October 27, 2026

1 Purpose

The estimator receives occasional whale-position measurements formed by triangulating two vehicle bearings. It combines each measured position with a motion model to estimate the whale's north/east position and velocity. Odyssey projects the latest estimate forward between measurements so a vehicle can steer toward the predicted whale location.

The ROS whale-estimator node runs a Kalman filter when a new triangulated position arrives. Odyssey then predicts where the whale would be at the current control time using the latest estimate and the same constant-rate return model.

The current configuration sets the north baseline velocity to 1.5 m/s, the east baseline to 0 m/s, and the velocity return time to 1800 s. Therefore the model does not bring the whole velocity vector to zero: north tends toward 1.5 m/s, while east tends toward zero. This is an estimator assumption; it does not change the whale simulator's actual trajectory.

2 State for one axis

North and east are filtered independently using the same equations. For one axis, the state is

$$\mathbf{x} = \begin{bmatrix} p \\ v \end{bmatrix},$$

where p is position in metres and v is velocity in metres per second. The filter also stores

$$\mathbf{P} = \begin{bmatrix} P_{pp} & P_{pv} \\ P_{pv} & P_{vv} \end{bmatrix}.$$

The diagonal entries are position and velocity variances. The off-diagonal entry is their covariance, describing how position and velocity errors vary together. A triangulated position supplies measured position z and measurement variance R; it does not directly measure velocity.

3 Motion model: constant-rate return toward a baseline

Let v_0 be the configured baseline velocity and τ the return duration. At the start of a prediction interval, let v_s be the estimated velocity and $d_0 = v_s - v_0$ its difference from baseline. The model removes that starting difference at a constant rate, reaching baseline after τ seconds if no newer position estimate arrives:

$$v(t) = v_0 + d_0 \max(1 - t/\tau, 0).$$

For $0 \le t < \tau$, velocity changes along a straight line. At and after τ, it remains at v_0. For a given prediction, the rate is $|d_0|/\tau$; a new position estimate starts a new prediction from its updated velocity.

Position is the integral of that changing velocity:

$$p(t) = \begin{cases} p(0) + v_0 t + d_0 \left(t - \dfrac{t^2}{2\tau} \right), & 0 \le t < \tau, \\ p(0) + v_0 t + d_0 \dfrac{\tau}{2}, & t \ge \tau. \end{cases}$$

This keeps position continuous and accounts for travel while velocity returns to baseline. Unlike an exponential return, the linear rule reaches baseline in finite time.

4 Discrete Kalman prediction

For a measurement interval Δt, define the remaining fraction of the starting velocity difference and the integrated travel factor:

$$a = \max(1 - \Delta t/\tau, 0), \qquad T = \begin{cases} \Delta t - \dfrac{\Delta t^2}{2\tau}, & \Delta t < \tau, \\ \tau/2, & \Delta t \ge \tau. \end{cases}$$

The code calls T `travel_time`. The state prediction is

$$\mathbf{x}^- = \mathbf{F}\mathbf{x} + \mathbf{B}\mathbf{u}, \qquad \mathbf{F} = \begin{bmatrix} 1 & T \\ 0 & a \end{bmatrix}, \qquad \mathbf{B}\mathbf{u} = \begin{bmatrix} v_0(\Delta t - T) \\ v_0(1 - a) \end{bmatrix}.$$

F propagates position and velocity deviations; the control term adds back baseline motion. Equivalently,

$$p^- = p + v_0 \Delta t + (v - v_0)T, \qquad v^- = v_0 + (v - v_0)a.$$

If the interval lasts at least τ, the starting velocity difference has fully returned to baseline.

5 Process noise: origin of Q

The process-noise covariance **Q** describes uncertainty added by random acceleration during one prediction interval. The implementation uses white acceleration noise with intensity q. Integrating that noise for a constant-velocity state over Δt gives

$$Q_{pp} = q\frac{\Delta t^3}{3}, \qquad Q_{pv} = q\frac{\Delta t^2}{2}, \qquad Q_{vv} = q\Delta t.$$

Therefore,

$$\mathbf{Q} = \begin{bmatrix} Q_{pp} & Q_{pv} \\ Q_{pv} & Q_{vv} \end{bmatrix}.$$

These values add uncertainty in position, the relationship between position and velocity, and velocity. The deterministic prediction handles the constant-rate return toward baseline; q describes random acceleration around that prediction. For simplicity, this Q calculation treats the random acceleration as acting on a constant-velocity state during the interval; it is an approximation alongside the deterministic return model.

6 Measurement correction

A position measurement observes only position:

$$z = \mathbf{H}\mathbf{x} + \text{measurement noise}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

The filter predicts the state and its uncertainty:

$$\mathbf{x}^- = \mathbf{F}\mathbf{x} + \mathbf{B}\mathbf{u}, \qquad \mathbf{P}^- = \mathbf{F}\mathbf{P}\mathbf{F}^T + \mathbf{Q}.$$

It compares measured and predicted position. Their difference, the innovation, and its variance are

$$y = z - \mathbf{H}\mathbf{x}^-, \qquad S = \mathbf{H}\mathbf{P}^-\mathbf{H}^T + R.$$

The Kalman gain determines how strongly the filter applies that difference:

$$\mathbf{K} = \mathbf{P}^-\mathbf{H}^T S^{-1}.$$

The corrected state and covariance are

$$\mathbf{x} = \mathbf{x}^- + \mathbf{K}y, \qquad \mathbf{P} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}^-.$$

Smaller R means the measurement is trusted more; larger R means the prediction is trusted more. A position correction can also change velocity because $\mathbf{P}$ tracks position–velocity covariance.

For the first position, the estimator sets position to the measured value and velocity to the configured baseline. The position variance comes from that measurement; initial velocity variance comes from `initial_velocity_variance_m2ps2`.

7 Velocity fit from recent positions

The estimator also fits a straight line through recent triangulated positions to estimate velocity. It waits for at least three positions, uses at most the latest five, and removes positions older than `velocity_window_max_age_s`. The slope against simulated time is

$$\hat{v} = \frac{\sum_i (t_i - \bar{t})(p_i - \bar{p})}{\sum_i (t_i - \bar{t})^2}.$$

North and east slopes are computed separately. Fit residuals contribute to the velocity variance, and a term involving q allows for acceleration during the fitting interval.

The Kalman filter has already used these same positions. Therefore the fitted velocity replaces the current velocity estimate and variance; it is not treated as a new independent Kalman measurement. The code also sets the position–velocity covariance to zero after applying the fit.

8 Odyssey prediction between measurements

Odyssey projects the latest estimate to the current simulated time using the same constant-rate return:

$$\hat{p}(t) = \begin{cases} \hat{p}(0) + v_0 t + (\hat{v}(0) - v_0)\left(t - \dfrac{t^2}{2\tau}\right), & t < \tau, \\ \hat{p}(0) + v_0 t + (\hat{v}(0) - v_0)\dfrac{\tau}{2}, & t \geq \tau, \end{cases}$$

$$\hat{v}(t) = v_0 + (\hat{v}(0) - v_0)\max(1 - t/\tau, 0).$$

The trailing behavior steers toward the projected position. Under the current configuration, projected north velocity approaches 1.5 m/s and east velocity approaches zero. If both baseline components were zero, the estimated velocity vector would gradually approach zero instead.

9 Configuration values

Parameter	Meaning
`q`	Acceleration-noise intensity. Larger values add more uncertainty between measurements. Units: m^2/s^3.
`initial_velocity_variance_m2ps2`	Uncertainty in starting velocity; separate from q.
`whale_velocity_return_time_s`	Time for the starting velocity difference to return linearly to baseline.
`whale_baseline_velocity_north_mps`, `whale_baseline_velocity_east_mps`	Long-term north/east velocity assumptions.
`velocity_window_max_age_s`	Maximum age of positions in the recent-position velocity fit.
R	Variance of an individual triangulated position; controls how strongly it affects the Kalman update.

10 Assumptions and limitations

The model assumes smooth velocity changes around a configured baseline with Gaussian acceleration noise. The baseline is a prediction assumption, not a command to the simulated whale. North and east are filtered independently, so their motion correlation is not represented. The recent-position velocity fit uses the same observations as the Kalman filter and overwrites the velocity estimate rather than adding an independent observation.

References

[1] D. S. Johnson, J. M. London, M.-A. Lea, and J. W. Durban, "Continuous-time correlated random walk model for animal telemetry data," *Ecology*, vol. 89, no. 5, pp. 1208–1215, 2008. doi:10.1890/07-1032.1.